# Estimation Methods for Regulatory Inheritance Probability in Yeast Gene Duplication

**Project:** SGD Yeast GRN — Scruse et al. (2024) Inheritance Probability Model
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## Background

When a gene duplicates in *Saccharomyces cerevisiae*, the new copy may or may not inherit the regulatory relationships of the original — that is, whether the transcription factors (TFs) that bound to the original gene's promoter will also bind to the duplicate's promoter. The **inheritance probability** $\pi$ (pi) quantifies this: $\pi = 1$ means every regulatory link is always passed on; $\pi = 0$ means none ever are.

In a network with $k$ gene families involved in a regulatory subnetwork motif, each family $i$ has its own inheritance probability $\pi_i$. The vector $\vec{\pi} = (\pi_1, \pi_2, \ldots, \pi_k)$ describes the full picture. The sum $\hat{\pi} = \sum \pi_i$ is a key aggregate used in the mathematical model (Theorem 4 of Scruse, Arnold & Robinson, 2024) to compute the expected number of motif instances in the regulatory network.

This document describes all seven estimation methods plus the multi-signal ensemble approach, where each method comes from, how it is computed, and how accurate each one is for different estimation goals.

## The Two Estimation Goals

It is important to distinguish between two related but different questions:

1. **What is the aggregate $\hat{\pi} = \sum \pi_i$?**
   This controls the expected motif count and significance test. A single number capturing the total inheritance probability across all families in the motif.

2. **What is each family's individual $\pi_i$?**
   This tells you which families carry more or less of the inheritance signal. Required for understanding family-specific regulatory conservation patterns.

Not all methods can answer both questions. Method 2 is the only approach that directly solves for the aggregate $\hat{\pi}$ from network data — but it says nothing about how $\hat{\pi}$ is distributed across families. Methods 1, 3, 4, 5, 6, and 7 each produce per-family estimates from independent biological signals.

## Key Terms and Notation

The methods below share a common set of symbols and terms. This section defines each one for reference.

### Core symbols

| Symbol | Meaning |
|---|---|
| $\pi_i$ | Inheritance probability for gene family $i$ — the probability that family $i$'s regulatory links are preserved after gene duplication. Ranges from 0 (never inherited) to 1 (always inherited). |
| $\hat{\pi}$ | The *aggregate* inheritance probability, $\hat{\pi} = \sum \pi_i$, summed across all $k$ families in a motif. This is the single number that drives the Theorem 4 expected motif count and the significance test. |
| $\vec{\pi}$ | The inheritance probability *vector* $(\pi_1, \pi_2, \ldots, \pi_k)$, i.e. the full per-family breakdown rather than just the sum. |
| $\pi_2$ | The Method 6 (protein sequence homology) estimate: percent pairwise protein identity divided by 100. |
| $\pi_3$ | The Method 5 (TFBS conservation) estimate: the fraction of Y1000+ genomes in which a TF's binding site is retained at the orthologous promoter. |
| $\pi_4$ | The Method 7 (IC-weighted SNP rate) estimate: one minus the information-content-weighted polymorphism rate at a TF's binding-site positions. |
| $k$ | The number of gene families involved in a given regulatory subnetwork motif. |
| $m$ | The number of founding gene families before duplication (a Theorem 4 model parameter). |
| $n$ | The total gene count after duplication (a Theorem 4 model parameter). |
| $\|M(n)\|$ | The observed motif count — the number of times the chosen $k$-family regulatory subnetwork pattern appears in the *S. cerevisiae* TF network. |
| $\Gamma$ | The gamma function, used in Theorem 4's expected motif count formula: $E[\|M(n)\|] = \dfrac{\Gamma(\hat{\pi}+n)\,\Gamma(m)}{\Gamma(\hat{\pi}+m)\,\Gamma(n)}$. |
| $\sigma$ | Standard deviation. In the ensemble method, $\pm\sigma$ across the per-method signals for a family measures how much the methods agree with one another. |

## Method-specific terms

| Term | Meaning |
|---|---|
| $e_i$ | The raw evidence score for gene family $i$, computed as the two-level mean of SGD evidence-code reliability weights: first averaging over codes within each TF, then averaging over TFs within the family (see Equation 1). This is the base quantity that Method 1 adjusts to produce $\pi_i^{(1)}$. |
| $F_i$ | The set of TFs belonging to gene family $i$, as determined by the chosen family-grouping method. $\|F_i\|$ is the number of TFs in that set, and the sum in Equation 1 runs over $g \in F_i$. |

| Term | Meaning |
|---|---|
| $C_g$ | The set of SGD GO evidence codes annotated to TF $g$. If a TF has multiple codes, $\|C_g\| > 1$ and the inner sum in Equation 1 averages over all of them. |
| $q(c)$ | The fixed reliability weight of evidence code $c$, $q(c) \in [0, 1]$. Maps each SGD/GO evidence type to a numerical prior: e.g. $q(\text{IDA}) = 0.90$, $q(\text{IEA}) = 0.30$, $q(\text{ND}) = 0.10$. The full mapping is given in the Method 1 evidence-code table. |
| $\delta_i^{\text{db}}$ | Indicator variable (Method 1): equals 1 if the majority of TFs in family $i$ carry a confirmed DNA-binding domain annotation; 0 otherwise. Applied as a +0.05 additive correction to $e_i$ in Equation 2. Reflects the empirical observation that confirmed DNA-binding roles correlate with regulatory-link conservation after duplication. |
| $\delta_i^{\text{act}}$ | Indicator variable (Method 1): equals 1 if the majority of TFs in family $i$ are annotated as transcriptional activators; 0 otherwise. Applied as a +0.03 additive correction to $e_i$ in Equation 2. Reflects the empirical observation that activator annotations correlate with regulatory-link conservation after duplication. |
| $w_i$ | Allocation weight for family $i$, used *only* in Method 2 to distribute the aggregate $\hat{\pi}$ across families after it has been recovered from Theorem 4: $\pi_i = \hat{\pi} \times \dfrac{w_i}{\sum_j w_j}$. These weights are typically taken from Method 1's evidence scores and are an assumption, not something derived from network data. |
| $\text{factor}_i$ | The Method 4 consensus (binding-flexibility) multiplier for TF $i$, always $\geq 1.0$. Derived from the number, mean length, and mean ambiguity of a TF's YEASTRACT IUPAC consensus sequences; a higher factor means a more sequence-tolerant (promiscuous) binder. |
| $IC_p$ | Information content of position $p$ in a TF's position weight matrix (PWM), computed as $IC_p = 2 + \sum_b f_{bp} \log_2 f_{bp}$ (summed over bases A, C, G, T). High IC marks a functionally critical, highly conserved position; low IC marks a degenerate, tolerant position. |
| $\text{poly\_rate}_p$ | Polymorphism rate at PWM position $p$ — the fraction of Y1000+ species whose nucleotide at that position differs from the *S. cerevisiae* reference. |
| weighted_poly | The IC-weighted average polymorphism rate across all positions in a binding site, used to compute $\pi_4$ in Method 7 ($\pi_4 = 1 - \text{weighted\_poly}$). |
| $\mu$ | The cross-strain mean in Method 3 ($\mu \approx 0.59$) — the average SNP-divergence-based inheritance proxy across the 11 sequenced YFL039C strains, applied uniformly to every family. |
| $A_i$ | In the ensemble method, the set of methods that produced a valid $\pi_i$ estimate for family $i$. The ensemble mean is the unweighted average over this set: $\pi_i^{\text{ensemble}} = \dfrac{1}{\|A_i\|} \sum_{j \in A_i} \pi_i^{(j)}$. |

| Term | Meaning |
|---|---|
| Evidence code | An SGD/GO annotation (e.g. IDA, IMP, IEA) that records how a TF→gene regulatory relationship was experimentally or computationally established. Method 1 maps each code to a fixed weight $q(c)$ between 0.10 and 0.90. |
| PWM / PFM | Position weight matrix / position frequency matrix — a per-position probabilistic model of a TF's DNA-binding preference, sourced from JASPAR 2024 and used to scan promoter sequences for binding-site matches (Methods 5 and 7). |
| Ortholog | The corresponding gene in another species' genome, identified across the 1,154 Y1000+ yeast genomes to assess cross-species conservation of a binding site (Methods 5 and 7). |

## Method 1 — Evidence-Based (SGD Experimental Evidence Codes)

### What it estimates

Per-family $\pi_i$, derived from the quality of experimental evidence confirming each TF's regulatory role in *S. cerevisiae*.

### Data source

`sgd_transcription_factors.csv` — compiled from the *Saccharomyces* Genome Database (SGD). Each TF→gene regulatory relationship carries one or more GO evidence codes, recorded by curators as they add experimental results to the database.

### How it works

Each evidence code $c$ is assigned a reliability weight $q(c) \in [0, 1]$ based on how directly it demonstrates biological function:

| Evidence Code | Type | $q(c)$ |
|---|---|---|
| IDA | Inferred from Direct Assay | 0.90 |
| IMP | Inferred from Mutant Phenotype | 0.82 |
| IGI | Inferred from Genetic Interaction | 0.72 |
| IPI | Inferred from Physical Interaction | 0.68 |
| IBA | Inferred from Biological Aspect of Ancestor | 0.58 |
| HDA | High-throughput Direct Assay | 0.55 |
| IC | Inferred by Curator | 0.50 |
| RCA | Reviewed Computational Analysis | 0.45 |
| TAS | Traceable Author Statement | 0.35 |
| IEA | Inferred from Electronic Annotation (automated) | 0.30 |
| NAS | Non-traceable Author Statement | 0.20 |
| ND | No biological Data available | 0.10 |

The raw evidence score for family $i$ is computed as a two-level mean: first averaging over all codes $C_g$ annotated to each TF $g$, then averaging over all TFs $F_i$ in the family:

$$e_i = \frac{1}{|F_i|} \sum_{g \in F_i} \frac{1}{|C_g|} \sum_{c \in C_g} q(c). \tag{1}$$

Two small indicator corrections are then applied to account for structural properties that correlate empirically with regulatory-link conservation:

$$\pi_i^{(1)} = \min\Big(1,\ e_i + 0.05\,\delta_i^{\text{db}} + 0.03\,\delta_i^{\text{act}}\Big)\,, \tag{2}$$

where $\delta_i^{\text{db}} = 1$ if the majority of TFs in $F_i$ carry a confirmed DNA-binding domain annotation (0 otherwise), and $\delta_i^{\text{act}} = 1$ if the majority are annotated as transcriptional activators. Because GO term annotations do not resolve to TF-level experimental detail in the current pipeline, all seven families typically receive $e_i$ scores in a narrow range ($e_i \approx 0.54$–$0.64$), yielding nearly uniform per-family estimates before the $\delta$ corrections are applied.

### Strengths

- Instantly available for all 179 TFs in the dataset; no external computation required.
- Biologically grounded: a TF confirmed by direct biochemical assay (IDA, $q = 0.90$) really is more likely to have stable regulatory links than one annotated purely by automated sequence similarity (IEA, $q = 0.30$).
- The $\delta$ corrections add biologically motivated differentiation beyond the raw evidence mean: confirmed DNA-binding domains and activator roles both correlate with regulatory-link conservation after duplication.
- Produces genuinely different estimates per family based on experimental track record and TF structural properties.

### Limitations

- **Narrow effective range.** The evidence codes span $q(c) \in [0.10, 0.90]$, so $e_i$ is constrained to roughly 0.31–0.77 for typical families. The $\delta$ corrections add at most +0.08, extending the ceiling to $\sim$0.85 but leaving very low inheritance families systematically over-estimated and the full $[0, 1]$ range unreachable.
- **No network or evolutionary data:** it reflects how well-studied a TF is and whether it is structurally a DNA-binder or activator, not how stable its regulatory links actually are across duplication events or evolutionary time.
- Families with sparse SGD curation receive conservative low-quality defaults regardless of their true inheritance pattern.

# Method 2 — Moment Estimation via Theorem 4 (MLE/Theorem 4 Inversion)

### What it estimates

The **aggregate** $\hat{\pi} = \sum \pi_i$ only. Does **not** produce per-family $\pi_i$ values.

### Data source

The observed motif count $|M(n)|$ — the number of times the chosen $k$-family regulatory subnetwork pattern appears in the *S. cerevisiae* TF network, combined with the model parameters $m$ (number of founding gene families) and $n$ (total gene count after duplication).

### How it works

Scruse et al. Theorem 4 gives the expected number of motif instances under Partial Duplication:

$$E[|M(n)|] = \frac{\Gamma(\hat{\pi} + n)\,\Gamma(m)}{\Gamma(\hat{\pi} + m)\,\Gamma(n)}$$

where $\hat{\pi} = \sum \pi_i$ is the total inheritance probability and $\Gamma$ is the gamma function. The method inverts this equation numerically (Brent root-finding) given the observed motif count, recovering $\hat{\pi}$ exactly.

Because Theorem 4 depends only on $\hat{\pi}$ and not on individual $\pi_i$ values, once $\hat{\pi}$ is found the code distributes it proportionally across families using weights from another method (typically Method 1 evidence scores):

$$\pi_i = \hat{\pi} \times \frac{w_i}{\sum_j w_j}$$

This allocation step is an assumption, not a derivation from the data.

### Strengths

- The most mathematically rigorous method for estimating the aggregate $\hat{\pi}$.
- Direct connection to the Scruse et al. statistical test: the significance $z$-score and $p$-value in the Motif Significance tab are computed from $\hat{\pi}$, so this method provides the most internally consistent estimate for that test.
- Works with any observed motif count; no external biological data required.

### Limitations

- **Cannot distinguish per-family $\pi_i$ values.** This is a fundamental mathematical constraint: Theorem 4 is a function of $\sum \pi_i$ only. Infinitely many different per-family distributions produce the same $\hat{\pi}$.
- The "observed count" in the current implementation uses the Cartesian-product upper bound ($c_1 \times c_2 \times \cdots \times c_k$) rather than a true count of network instances, which limits the precision of the inversion.
- Sensitive to the choice of allocation weights: once $\hat{\pi}$ is allocated across families using external weights, the per-family values are only as good as those weights.

# Method 3 — SNP Divergence Proxy (YFL039C Strain Variation)

## What it estimates

A scalar inheritance probability derived from sequence divergence, applied uniformly to all gene families. Per-family differentiation is only possible if the selected genes happen to appear in the YFL039C dataset (rare in practice).

## Data source

`sgd_YFL039C_snps.csv` and `sgd_YFL039C_inheritance_vectors.csv` — SNP data from 11 sequenced strains of gene YFL039C (a well-studied *S. cerevisiae* locus used here as a calibration target). For each strain, the percentage of alternative alleles at that locus is recorded.

## How it works

For each sequenced strain $s$, the inheritance proxy is:

$$\pi_s = 1 - \frac{\text{pct\_alt}_s}{100}$$

The cross-strain mean $\mu \approx 0.59$ is then used as the per-family estimate for all $k$ families in the motif.

The logic is: greater sequence divergence from the reference implies an older regulatory link. Older links have had more opportunity to be disrupted. Therefore, higher divergence $\rightarrow$ lower $\pi$.

## Strengths

- Computationally instant; no external genome-scale data needed.
- Provides a biologically motivated central estimate when no other information is available.
- Independent of the evidence-code framework (Methods 1, 4): uses a purely sequence-based signal.

## Limitations

- **No per-family differentiation.** All families receive the same $\mu \approx 0.59$, regardless of their actual conservation patterns. Method 3 is structurally identical to Method 2's uniform allocation in this respect.
- **Single-gene calibration.** YFL039C is one gene. Its divergence pattern may not represent the TFs of interest.
- **MSE is irreducible for heterogeneous profiles.** If the true inheritance probabilities differ across families, the mean-constant estimate will be wrong for every family simultaneously. The error equals the variance of the true profile around $\mu$.

# Method 4 — Consensus-Adjusted (YEASTRACT Binding Flexibility)

### What it estimates

Per-family $\pi_i$, starting from Method 1's evidence-based prior and adjusting upward based on how flexible each TF's binding sequence is.

### Data source

`yeastract_consensus.csv` — IUPAC consensus binding sequences from the YEASTRACT database (Monteiro et al., 2020), which curates experimentally confirmed TF binding data for *S. cerevisiae*. Each TF may have multiple consensus sequences of varying lengths and ambiguity levels.

### How it works

For each TF, a consensus factor is computed from:

- Number of distinct IUPAC consensus sequences the TF has
- Mean length of those sequences
- Mean ambiguity fraction (proportion of positions using degenerate IUPAC codes: N, R, Y, W, S, etc.)

A TF with many ambiguous binding sequences is a promiscuous binder: its binding site can tolerate sequence drift after duplication, making it more likely to retain its regulatory link in the duplicate gene's promoter. The factor is always $\geq 1.0$ (a minimum upward adjustment). The adjusted $\pi$ is:

$$\pi_i^{\text{adjusted}} = \min\Big(1.0,\ \pi_i^{(1)} \times \text{factor}_i\Big)$$

where $\pi_i^{(1)}$ is the Method 1 estimate from Equation 2. In practice, factors range from approximately 1.02 to 1.12.

### Strengths

- Adds biologically motivated information: binding specificity affects inheritance likelihood.
- Per-family differentiation: different TFs get different adjustment factors based on their actual binding sequence diversity.
- Always at least as large as Method 1 for the same gene, providing a consistent upper bound.

### Limitations

- **Narrow adjustment range (1.02–1.12).** Because YEASTRACT consensus sequences are generally quite specific, the flexibility signal is modest. Method 4 sits only slightly above Method 1 in practice and inherits the same structural ceiling ($\sim$0.85) and floor ($\sim$0.37).
- **Only an upward adjustment.** Method 4 cannot produce a lower estimate than Method 1 for the same TF, even when tighter binding would argue for lower inheritance.
- Requires YEASTRACT consensus data; TFs not in the database receive a default factor of 1.0 (making Method 4 identical to Method 1 for those genes).

## Method 5 — TFBS Conservation Across 1,154 Yeast Genomes ($\pi_3$)

### What it estimates

Per-family $\pi_i$, derived from how often each TF's binding site is retained in the upstream region of the orthologous gene across 1,154 yeast species spanning ∼1 billion years of evolution.

### Data source

**Y1000+ dataset** (Opulente et al., 2024, *Science* 384) — genomic assemblies, GFF3 gene annotations, and protein sequences for 1,154 *Saccharomycotina* yeast species, covering the most comprehensive yeast phylogenomic survey available.

**JASPAR 2024 Core Yeast TF collection** — 177 position frequency matrices (PFMs) for experimentally validated *S. cerevisiae* TF binding sites. Each PFM is scanned against the 1,000 bp upstream of the translational start site (promoter window) of the orthologous gene in each of the 1,154 genomes using position weight matrices (PWMs) with a significance threshold.

### How it works

For each TF→gene regulatory edge in the *S. cerevisiae* network:

1. Find the orthologous gene in each of the 1,154 Y1000+ genomes.
2. Extract the 1,000 bp upstream of its translational start.
3. Scan that sequence with the TF's JASPAR 2024 PWM.
4. Record whether the scan produces a significant hit (above a log-odds threshold).
5. Compute the retention fraction: (number of genomes with a significant hit) / (number of genomes where the ortholog was found).

This retention fraction is $\pi_3$ for that TF→gene edge. The per-family estimate is the mean $\pi_3$ across all target genes of that TF.

### Strengths

- **Direct evolutionary evidence.** This is not a proxy — it actually measures whether the binding site survived across independent evolutionary replicates.
- **No structural floor or ceiling.** Unlike Methods 1 and 4, $\pi_3$ can in principle reach any value from 0 to 1.
- **Per-family differentiation.** A TF conserved across most yeasts gets a high $\pi_i$; a lineage-specific TF gets a low one.
- **Largest biological dataset:** 1,154 genomes spanning ∼1 billion years of evolution provides statistical power no single-species analysis can match.

### Limitations

- Requires substantial computation on first run (15–30 minutes to extract and scan all 1,154 genomes; cached afterward).
- PWM coverage. Only TFs with a JASPAR 2024 PWM can be assessed. TFs not in JASPAR have no $\pi_3$ estimate.

- Ortholog availability. The fraction is computed only over genomes where the ortholog was found. Genes with poor cross-species synteny conservation yield noisier estimates.
- PWM sensitivity threshold. The significance threshold affects which hits count. A conservative threshold will underestimate retention; a permissive one will overestimate it.

# Method 6 — Protein Sequence Homology ($\pi_2$)

## What it estimates

Per-family $\pi_i$ based on how similar each TF's protein sequence is to the other TFs in the dataset — a proxy for how closely related the paralogs are.

## Data source

**Y1000+ protein sequences** — protein FASTA files from the Y1000+ dataset, used to compute pairwise sequence identity between the 179 TFs using k-mer alignment over 8,001 TF pairs. Results are stored in `pi2_sequence_homology.csv`.

## How it works

For each pair of TFs, the percent protein sequence identity is computed using fast k-mer alignment. This is converted directly to a $\pi_2$ estimate:

$$\pi_2 = \frac{\text{pct\_identity}}{100}$$

For the per-family ensemble (and per-TF analysis in the Method Estimation Test), each TF's $\pi_2$ is its mean pairwise identity with all other TFs in the dataset — giving each TF a distinct value rather than a single mean across a selected subset.

TFs can also be clustered at 30%, 50%, or 80% identity thresholds to define gene families for use in the main model parameters.

## Strengths

- Protein-level similarity is a meaningful proxy: paralogs that are more similar at the protein level are more likely to share binding domains and regulatory targets.
- Fast to compute; runs in ~2 seconds regardless of which TFs are selected.
- Produces per-TF values when computed as per-TF mean identity.

## Limitations

- **One step removed from binding.** Protein similarity does not guarantee binding-site conservation. Two highly similar TFs may bind completely different target genes if their DNA-binding domain differs even slightly.
- **No evolutionary time-course.** Protein identity is a snapshot; it does not distinguish a recent duplication (high identity, links not yet diverged) from an ancient one that has converged to high identity through selection.
- **Indirect link to regulatory inheritance.** The $\pi$ in the Scruse model describes binding-

site inheritance specifically, while protein identity describes protein-sequence inheritance more broadly.

## Method 7 — IC-Weighted SNP Rate at Binding Sites ($\pi_4$)

### What it estimates

Per-family $\pi_i$ based on how many mutations have accumulated at the exact positions within each TF's binding sites across the Y1000+ yeast species.

### Data source

**Y1000+ genomic sequences and JASPAR 2024 PWMs** — the same upstream sequence scanning infrastructure as Method 5, but instead of recording whether the hit exists, it records the nucleotide identity at every position of the binding site across species.

Results are stored in `pi4_snp_binding_sites.csv`.

### How it works

For each position $p$ in a TF's PWM:

1. Compute the **information content (IC)** of that position from the PWM:

   $$IC_p = 2 + \sum_{b \in \{A,C,G,T\}} f_{bp} \log_2 f_{bp}$$

   High IC means the position is critical (very conserved in the known binding motif). Low IC means the position is degenerate (TF tolerates any nucleotide there).

2. Across all Y1000+ species where the ortholog was found, compute the **polymorphism rate** at position $p$: fraction of species with a different nucleotide than the reference.

3. Compute the IC-weighted polymorphism rate:

   $$\text{weighted\_poly} = \frac{\sum_p IC_p \times \text{poly\_rate}_p}{\sum_p IC_p}$$

4. The $\pi_4$ estimate is:

   $$\pi_4 = 1 - \text{weighted\_poly}$$

The IC-weighting means that a mutation at a highly conserved, critical position of the binding site contributes much more to the polymorphism score than a mutation at a degenerate position.

### Strengths

- **Most mechanistically precise.** This measures whether the binding site itself has changed, not just whether it produces a significant hit. A site that has drifted subtly — still passing the PWM threshold but accumulating mutations at critical positions — is correctly identified as deteriorating.

- **Information-content weighting** ensures that mutations at functionally critical positions dominate the signal, filtering out noise from degenerate positions.

- Per-family differentiation: different TFs have different patterns of binding-site conservation.

### Limitations

- Same infrastructure requirements as Method 5 (Y1000+ genomes, JASPAR PWMs, 15–30 min first-run computation).
- **Requires a significant binding hit first.** $\pi_4$ is computed at the binding-site positions, so it can only be calculated where a significant PWM hit was found. Genes with low retention (low $\pi_3$) have fewer binding hits, making the $\pi_4$ estimate noisier.
- **Correlated with Method 5.** Because both use the same Y1000+ scan, their estimates are not fully independent; a TF with high TFBS conservation (high $\pi_3$) will tend to also have low binding-site polymorphism (high $\pi_4$). The two signals are complementary rather than redundant, but they share a common data source.

## Multi-Signal Ensemble — Per-Family Combination of All Methods

### What it estimates

Per-family $\pi_i$ for each gene family, combining all six methods that produce independent per-family signals: Methods 1, 3, 4, 5, 6, and 7.

### Data source

All of the above — SGD evidence codes, YEASTRACT consensus sequences, Y1000+ genomes, and JASPAR 2024 PWMs. Method 2 is deliberately excluded.

### How it works

For each gene family $i$:

1. Collect the $\pi_i$ estimate from every available method (M1, M3, M4, M5, M6, M7).
2. Compute the unweighted mean of all non-missing values:
$$\pi_i^{\text{ensemble}} = \frac{1}{|\mathcal{A}_i|} \sum_{j \in \mathcal{A}_i} \pi_i^{(j)}$$
where $\mathcal{A}_i$ is the set of methods that produced a valid estimate for family $i$.
3. Also compute $\pm\sigma$ (standard deviation across the signals) as a measure of how much the methods agree.

The app displays the full per-signal breakdown alongside the ensemble mean, so the user can see exactly which signals drove a high or low estimate.

### Why Method 2 is excluded

Theorem 4 depends only on $\hat{\pi} = \sum \pi_i$. Inverting it recovers the total inheritance probability perfectly but provides no information about how that total is distributed across individual families. Including a uniform $\hat{\pi}/k$ allocation from Method 2 would pull every family's ensemble mean toward the same value, diluting the differentiation provided by the other methods.

### What conflicting signals mean

A large $\pm\sigma$ across signals for a given family is itself biologically informative:

- **High M1/M4 + Low M5/M7:** The TF is well-confirmed experimentally in *S. cerevisiae* but its binding sites are not conserved across yeasts. This suggests a recent regulatory link, or one that may be *S. cerevisiae*-specific.
- **Low M1 + High M5:** The TF is poorly-studied in SGD (automated annotation only) but its binding sites are deeply conserved. This is a candidate for re-investigation — the low evidence score may reflect curation lag rather than weak biology.
- **Uniform signals:** All methods agree, giving high confidence in the ensemble estimate.

### Strengths

- **Most robust per-family estimate available.** Averaging across independent signals reduces the impact of any single method's systematic errors.
- **Gracefully degrades.** When Y1000+ data is not available, the ensemble uses M1, M3, M4 and still provides differentiated estimates.
- **Per-signal breakdown** provides interpretability beyond the final number.

### Limitations

- **Unweighted averaging.** All methods contribute equally, even when some are more relevant. A future improvement would be to weight signals by their track record on held-out data.
- **M3 is quasi-constant.** In practice, most selected TFs will not be in the YFL039C strain dataset, so M3 contributes the same constant ($\mu \approx 0.59$) to every family, slightly pulling the ensemble toward that value.
- **Y1000+ dependency for full power.** Without M5, M6, and M7, the ensemble is constrained by the same evidence-code range as M1 (∼0.31–0.85).

## Accuracy Ranking

### For per-family $\pi_i$ estimation (recommended goal)

| Rank | Method | Why |
|---|---|---|
| 1 | **Multi-signal Ensemble** | Combines all six independent signals; reduces systematic errors from any single method; full power when Y1000+ data is available |
| 2 | **M5 — TFBS Conservation ($\pi_3$)** | Best single method; directly measures regulatory link retention across 1,154 independent evolutionary replicates; no structural range constraint |

| Rank | Method | Why |
|---|---|---|
| 3 | **M7 — Binding-site SNPs ($\pi_4$)** | Most mechanistically precise single signal; IC-weighting focuses on critical positions; complementary to M5 |
| 4 | **M6 — Sequence Homology ($\pi_2$)** | Useful per-TF differentiation; but protein similarity is one step removed from binding-site conservation |
| 5 | **M1 — Evidence-based** | Reliable prior with $\delta$ corrections for DNA-binding and activator roles; per-family differentiation; but narrow effective range |
| 6 | **M4 — Consensus-adjusted** | Small improvement over M1; same range problem; useful when YEASTRACT data available |
| 7 | **M3 — SNP Divergence** | Returns same constant for all families unless gene is in YFL039C dataset; no per-family information for typical TF selections |
| 8 | **M2 — Moment Estimation** | **Cannot produce per-family estimates.** Recovers $\hat{\pi}$ only; per-family allocation is assumed from weights, not derived from data |

### For aggregate $\hat{\pi} = \sum \pi_i$ estimation (significance testing)

| Rank | Method | Why |
|---|---|---|
| 1 | **M2 — Moment Estimation** | Mathematically exact; Theorem 4 inversion is the only approach that directly estimates $\hat{\pi}$ from network structure |
| 2 | **Multi-signal Ensemble** | Sum of per-family ensemble means; reasonable aggregate when M2 is not available or when per-family differences matter |
| 3 | **M5 — TFBS Conservation** | Data-rich aggregate; mean retention fraction across 1,154 species provides a biologically well-grounded total |
| 4–7 | M1, M4, M7, M6 | Progressively more indirect; all require external calibration relative to the motif-count model |
| Last | **M3 — SNP Divergence** | Single locus, single species; least informative about the actual network structure being tested |

## Summary Table

| Method | Data needed | Per-family? | Range | Best use case |
|---|---|---|---|---|
| M1 Evidence | SGD annotation | Yes | 0.10–0.85 | Prior when no other data available |
| M2 Moment Est. | Observed motif count | No ($\hat{\pi}$ only) | Unconstrained | Significance test; aggregate $\hat{\pi}$ |
| M3 SNP Divergence | YFL039C SNP file | Only if gene in dataset | ∼0.59 constant | Baseline when no per-gene data |
| M4 Consensus | YEASTRACT binding seqs | Yes | ∼0.37–0.85 | Refinement of M1 for YEASTRACT TFs |
| M5 TFBS Conserv. | Y1000+ + JASPAR PWMs | Yes | 0–1 | Best per-TF estimate for JASPAR TFs |
| M6 Seq. Homology | Y1000+ proteins | Yes | 0–1 | Protein-level family grouping |
| M7 Binding-site SNPs | Y1000+ + JASPAR PWMs | Yes | 0–1 | Mechanistic precision at binding positions |
| Ensemble | All of the above | Yes | Avg of available | Default per-family $\pi_i$ recommendation |

## Data Sources Referenced

| Source | Citation | What we use |
|---|---|---|
| SGD — Saccharomyces Genome Database | Cherry et al. (2012) | Evidence codes, TF annotations, GO terms |
| YEASTRACT | Monteiro et al. (2020) | IUPAC consensus binding sequences for 127 TFs |
| JASPAR 2024 | Castro-Mondragon et al. (2022) | 177 position frequency matrices for yeast TFs |
| Y1000+ | Opulente et al. (2024), *Science* 384 | 1,154 yeast genomes, GFF3 annotations, protein sequences |
| Scruse et al. (2024) | arXiv:2405.03148 | Mathematical framework: Theorems 1–8, Pólya urn model |

## Practical Recommendation

For any new analysis:

1. **Run the ensemble first.** It uses all available data and is the most balanced estimate of individual $\pi_i$ per family. Generate Y1000+ data first (one-time 15–30 min process) to unlock the three most powerful signals (M5, M6, M7).

2. **Use Method 2 for the significance test.** The Motif Significance tab uses $\hat{\pi} = \sum \pi_i$ as input to Theorem 4. Method 2 is the mathematically appropriate way to estimate that quantity from the observed network.

3. **Use the per-signal breakdown to flag interesting TFs.** When a TF's ensemble signals conflict strongly (large $\pm\sigma$), it is worth investigating individually — it may represent a recently acquired or lineage-specific regulatory link.

4. **Do not over-interpret Method 2's per-family output.** When Method 2 is used via the $\pi$ Estimator tab and distributes $\hat{\pi}$ proportionally, those per-family values reflect the allocation weights (evidence scores from Equation 2), not anything derived from the network. They are a starting point, not a result.